Reg. No. | | | | | | | | | | | [22MSCH1H01]

# FIRST SEMESTER UNIVERSITY EXAMINATION, JANUARY 2023

## INORGANIC CHEMISTRY-I

**Time: 03 HOURS** **Maximum Marks: 50**

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### SECTION – A

**1. Answer any FOUR of the following questions.** **4×5=20**

a. Discuss the structure of water with respect to hydrogen bonding interactions.

b. Sketch the molecular orbital energy level diagram for $N_2$ molecule and calculate the bond order.

c. Describe the crystallization process involved in making glass ceramics.

d. Discuss the properties of soft acids and soft bases with suitable examples.

e. Explain beta$^-$ and beta$^+$ decay processes with examples. Describe the relation between the nuclear radius and the mass number.

### SECTION – B

**2. Answer any TWO of the following questions.** **2×9=18**

a. Describe the salient features of VSEPR theory. Predict the shapes of $I_3^-$, $ClO_3F$, $ClF_3$, $PCl_5$ and $SF_4$ molecules.

b. i. List the postulates of molecular orbital theory.

ii. Explain the p-p and p-d combination of atomic orbitals.

c. List the following in order of increasing radius with reasons;

i. Na, $Na^+$, K, $Mg^{2+}$

ii. Na, Mg, Al.

### SECTION – C

**3. Answer the following question.** **12**

a. i. Explain in detail the classification of silicates on the basis of their structure. **04**

ii. Discuss the structure of $S_4N_4$ and $S_2N_2$ molecules. **04**

iii. What is a semi empirical mass formula? Explain the different terms used in it. **04**

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